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Achieving an Astonishing Milestone Through Innovative Approach and Project Execution of Unconventional Gas Field in Just 9 Months

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Abstract

Since IOC announced the Project Accelerate 100X back in 2022, many significant improvements and initiatives have been implemented to fast-track capital projects. The IOC 100X is more than just a transformation program; it is a bold commitment to achieving exponential performance across safety, sustainability, efficiency, and growth. It challenges every employee to think differently, act faster, and deliver smarter, pushing boundaries to unlock the full potential of our people and resources. Consequently, we took it upon ourselves to implement our unique 100X model into unconventional gas projects to both de-risk and accelerate gas developments. Given that the total identified reserves in Abu Dhabi amount reaching 160 trillion standard cubic feet (scf), IOC aimed to make unconventional gas meet 50% of the UAE's natural gas needs before 2030.

In this paper, we will exclusively discuss IOC's first unconventional gas project from the Onshore Field. This project deployed an exceptional project management model that integrated multiple acceleration approaches to bring the first commercial gas. This allowed us to complete the project in 9 months, approximately 3 times faster in terms of schedule and 4 times more cost-effective than standard EPC projects. Despite the highly compressed timeline, the project was delivered with outstanding quality, maintaining 100% Health, Safety, and Environment (HSE) compliance and achieving zero lost time incidents. This remarkable achievement underscores IOC's dedication to operational excellence and its ability to deliver complex projects efficiently while adhering to the highest safety standards.

The article will delve into the various project management strategies and execution approaches adopted for unconventional gas developments, with a specific focus on IOC's implementation model. It will highlight key takeaways and lessons learned from early execution phases, addressing both technical and non-technical challenges. In addition to showcasing project innovations and delivery milestones, the paper will also examine HSE integration from the early design stages through execution, including Emergency Planning Zone (EPZ) classification and stakeholder engagement. Challenges encountered ranging from procurement and constructions delays will be analyzed alongside the proactive mitigation measures undertaken. The paper concludes by outlining IOC's forward-looking strategies for scaling up unconventional developments efficiently, safely, and sustainably.

Introduction

The UC Onshore Concession Agreement was formally restated on 31st May 2023, reaffirming the partnership between IOC, which holds a 90% ownership stake, and TotalEnergies, which retains the remaining 10%. This strategic alliance is designed to foster close collaboration and harness the combined technical capabilities and operational experience of both parties to accelerate the development of a pilot-scale unconventional gas production facility. The project aims to produce and export 20 million standard cubic feet per day (MMSCFD) of commercial gas as a first-phase milestone.

Recognizing the unique challenges associated with unconventional resources, the project was not limited to technical innovation in drilling and multi-stage hydraulic fracturing. Instead, the scope was extended to adopt a transformative execution model that departs from traditional practices. The UC Onshore Project implemented a **Design-Build-Operate (DBO)** contracting strategy, which consolidates all project responsibilities ranging from conceptual design, detailed engineering, procurement, and construction (EPC) to commissioning and long-term operations and maintenance (O&M), under a single contractual umbrella managed by one contractor.

This integrated approach significantly improves project agility, coordination, and accountability. By eliminating the fragmented interfaces common in conventional models, the DBO framework enhances schedule adherence, cost control, and overall execution quality. It also ensures that design and operational considerations are seamlessly aligned from the outset, improving long-term asset performance.

The DBO model is particularly advantageous for unconventional developments still in the pilot phase, as it offers a flexible commercial structure that defers capital investment. Under this arrangement, performance guarantees are directly linked to production targets, and the full burden of operations and maintenance is transferred to the contractor. This structure significantly reduces the client's capital exposure during the early stages of development and aligns the contractor's incentives with long-term production performance. Figure 1 below compares the payment framework of the DBO model with that of a conventional EPC contract.

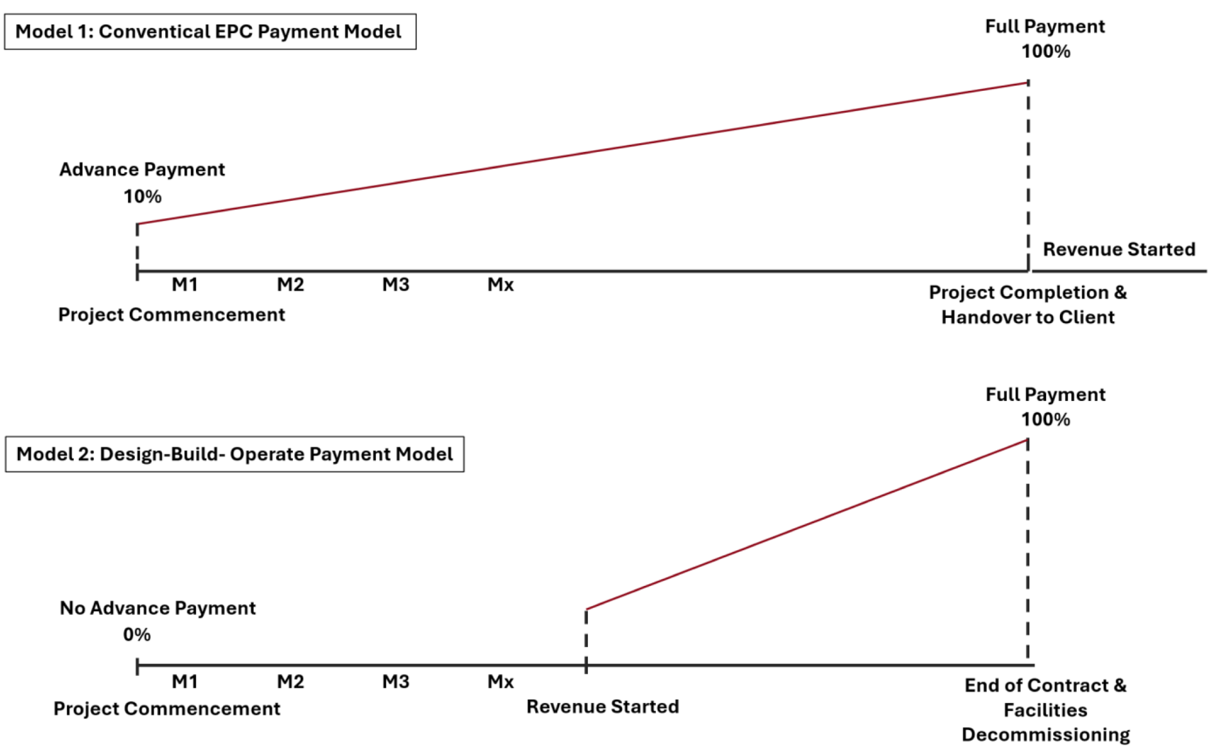

Figure 1—Comparison of Company Investment in EPC Payment Model and DBO Payment Model

In conventional EPC projects, the payment structure (finance model) is typically defined as either lump sum or cost reimbursable. In lump sum contracts, the client provides an advance payment commonly around 10% of the total contract value at the outset of the project. This advance enables the contractor to mobilize resources, establish site offices, initiate procurement of long-lead items, and cover other essential startup costs. Once mobilization is complete, the remainder of the payments is made progressively based on the achievement of pre-agreed milestones or the percentage of physical work completed. These milestones often correspond to engineering deliverables, equipment delivery, construction phases, systems commissioning, etc.

As the project progresses, payments are distributed incrementally throughout the execution lifecycle, culminating in the final payment upon successful commissioning and formal handover of the fully constructed and operational facility to the client. The client then takes full responsibility for operating and maintaining the asset. The EPC contractor's scope ends upon the delivery of a complete, permanent infrastructure that meets the agreed technical specifications and client company's standards.

Conversely, the DBO model introduces a fundamentally different financial and operational framework. Because the facilities are temporary, modular, and operated by the DBO contractor rather than the client, no advance payment is required. Instead, the contractor bears the upfront cost of facility construction and mobilization, recovering these costs only once the plant is online and delivering commercial production. Payments under the DBO model are structured as fixed, periodic fees based on an agreed production profile and runtime availability of the facility.

This outcome-based approach provides a natural alignment of incentives, as the contractor's revenue is directly tied to the operational performance and reliability of the facility rather than the physical completion of work. It places the commercial risk on the contractor, encouraging them to ensure high uptime, minimize downtime, and maintain steady production rates over the contract period. This model is particularly effective in pilot or early-phase developments, where the client seeks to minimize upfront capital expenses while still meeting strategic production objectives.

As illustrated in Figure 2, a comparative analysis is presented between the project execution timeline under a conventional EPC model and the accelerated schedule enabled by the DBO contracting strategy. A typical project executed through the conventional model progresses through five distinct phases: Assess, Select, Define, Execute, and Operate. The Assess phase, focused on feasibility studies and early-stage evaluations, generally spans 4 to 6 months. This is followed by the Select phase (Pre-FEED), which typically lasts 6 to 8 months, during which development concepts are matured, risks assessed, and cost estimates refined to a confidence level of ± 30%. The Define stage (FEED) can extend over 12 to 15 months, ensuring that the selected concept is engineered in sufficient detail for tendering and execution planning, in which the cost estimate confidence level is reduced to ± 10%. Finally, the Execute stage encompassing procurement, construction, and commissioning may take between 28 to 30 months depending on the scale and complexity of the project.

This linear and segmented process is designed to progressively build confidence in the business case by validating technical and commercial assumptions with increasing levels of detail. However, the conventional model is also inherently time-consuming due to the sequential nature of the phases, multiple tendering cycles, numerous internal and external stakeholder reviews, and repeated gate approvals each of which can cause significant delays.

In contrast, the DBO approach offers a radically streamlined alternative by collapsing all five phases into a single integrated execution model, managed by one contractor. This integration allows for concurrent planning and execution, enabling early engineering to overlap with procurement and site preparation. It also facilitates fast-track delivery of long-lead items and promotes iterative design flexibility throughout the project lifecycle. Moreover, the DBO contractor not only constructs the facility but also assumes responsibility for its commissioning, startup, and operation on behalf of IOC, eliminating handover delays and reducing overall schedule duration dramatically.

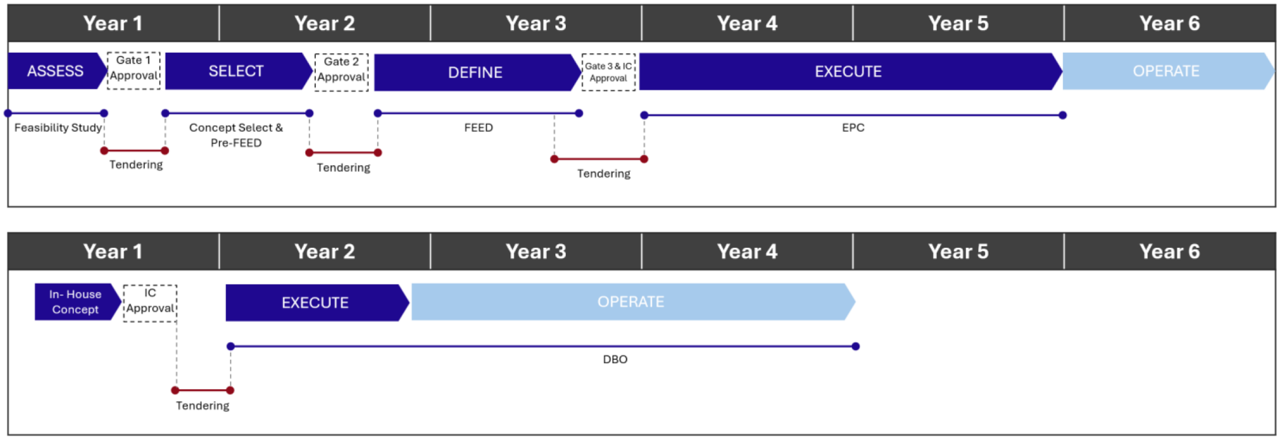

Figure 2—EPC Versus DBO Project Stages

DBO Execution Enablement: Front-End Alignment, Risk Mitigation, and Interface Readiness

A critical enabler of the project's success was the strategic and technically sound set of actions taken during the preparatory before contract award or site mobilization. The first and most pivotal step involved achieving early alignment with the subsurface team to define and freeze the project basis. The project concept was initiated based on the successful outcome of a single gas exploration well (Well-1). As such, all subsurface data including expected well count, production profiles, gas composition, water cut, flowing and shut-in pressures, and temperature was derived exclusively from this single data point.

Despite the limited dataset, this information provided a sufficient technical foundation for scoping the surface facilities. Notably, while Well-1 reported a minimal amount of H_2S, the team anticipated the possibility of increased levels during sustained production. Based on engineering assessment and previous field experience, the design envelope was expanded to accommodate higher H_2S concentration almost 3x more than the detected one. This foresight proved critical, as actual H_2S levels did rise post-startup. By embedding a conservative yet realistic design basis from the outset, the team avoided costly redesigns and late-stage modifications, underscoring the importance of front-end risk mitigation.

In the absence of formal Front-End Engineering Design (FEED) documentation, the project team developed sufficient engineering definition to enable competitive and informed bidding while retaining the DBO model's intended flexibility. Key scope elements such as hydraulic calculations for pipeline sizing were provided to ensure clarity for bidders. At the same time, bidders were intentionally given freedom to propose technology solutions in selected areas such as the choice between triethylene glycol (TEG) and molecular sieve dehydration systems, the number of process trains, and overall process flow configuration. This hybrid approach combining minimum technical definition with space for innovation stimulated technical creativity, enhanced competition, and resulted in optimized solutions tailored to the pilot nature of the project.

In another departure from conventional IOC project execution norms, bidders were allowed to propose international codes and standards were justified, rather than being restricted to IOC's internal technical guidelines. This pragmatic approach expanded the qualified bidder pool, reduced non-value-adding constraints, and preserved design integrity without compromising safety, operability, or reliability.

To mitigate procurement-related delays a common bottleneck in fast-track projects the team encouraged bidders to make use of available shelf or repurposed equipment. This included major components such as Gas Turbine Generators (GTGs), Emergency Diesel Generators (EDGs), and pressure vessels salvaged from terminated or dormant IOC projects. By reusing existing assets with verified integrity, the project reduced equipment lead times and expedited early mobilization.

In parallel with the technical scope development, the team initiated a conceptual Health, Safety, and Environmental Impact Assessment (HSEIA), recognizing the importance of early environmental clearance in maintaining schedule momentum. The proposed plant site was located in a greenfield area surrounded by sensitive receptors, including agricultural farms, a public highway, and the Al Houbara Protected Area. The Environment Agency – Abu Dhabi (EAD) did not recommend the initial site after consequence modeling revealed an unacceptable level of public risk in the event of a gas release impacting the nearby highway. Based on HSEIA findings, the team identified and secured a new site with lower risk exposure, which was communicated to all bidders preventing costly rework or relocation delays after contract award.

To enhance site understanding and facilitate more accurate construction planning, the project team completed and shared topographical and geotechnical surveys of the project area. While this data was not designated as rely-upon, it significantly improved bidders' ability to plan for site preparation, grading, and foundation design.

In a parallel effort to optimize existing infrastructure, the team identified a mothballed pipeline in the vicinity of the development area. After a thorough integrity assessment and the enhancement of its cathodic protection (CP) system, the asset was deemed suitable for reuse. Reintegrating this pipeline into the new network eliminated the need for constructing a new line, resulting in both capital savings and schedule gains.

Finally, early and structured engagement with IOC Gas was undertaken to define gas export interface requirements. This included alignment on tie-in location, pressure envelope, metering methodology, and control system integration. Addressing these interfaces proactively during the pre-award phase eliminated ambiguity and reduced the likelihood of scope creep or interface-related disruptions during execution.

Prioritizing Permitting and HSE Compliance for Accelerated DBO Execution

Once the DBO contractor was formally onboarded, one of the most immediate and critical priorities was the initiation of administrative processes related to permitting and regulatory compliance particularly the timely acquisition of No Objection Certificates (NOCs). These approvals were essential for enabling site access, land allocation, construction activities, and ultimately, authorization to build and operate a hydrocarbon facility.

For projects located within IOC's designated Concession Areas, these permitting processes are typically more streamlined due to the predefined land use agreements and IOC's established operational presence. This provides a level of administrative efficiency that accelerates project mobilization. However, the UC Onshore field presented a more complex case. Although its within IOC's broader domain, its location intersected with areas subject to heightened scrutiny and multi-agency oversight effectively introducing conditions different to those in IOC Concession Areas.

In such areas, the NOC process becomes significantly more rigorous. Numerous stakeholders and authorities must be consulted and aligned due to the potential for overlapping interests. This includes proximity to existing communities, natural reserves, critical infrastructure (such as communication cables or government facilities), public amenities, and long-term land development plans. For example, if government authorities were evaluating the future expansion of a nearby residential area, the project team would be required to demonstrate with documented technical evidence that the proposed facilities would not impose any risk or constraint to the planned community expansion. This includes potential environmental, health, and safety impacts.

In the case of the UC Onshore Project, a comprehensive combination of such sensitivities was present. Therefore, the project team immediately prioritized Health, Safety, and Environmental (HSE) studies, recognizing that early technical validation of site suitability was essential. The initial recommendations from the HSE Impact Assessment (HSEIA) called for the relocation of the Primary Processing Facilities (PPF) to minimize proximity risks to receptors and sensitive boundaries. This early action was fundamental as the project transitioned from a single-well exploration scope to a full-scale development plan. Temporary permits typically suitable for pilot or early-phase activities were no longer viable, and a robust long-term mitigation plan was necessary to ensure the safety of surrounding communities, assets, and personnel.

To address social and safety risks effectively, IOC's Social Risk Management Team (SRMT) played a leading role. Working in close collaboration with government stakeholders, the SRMT coordinated the notification and evacuation of impacted residents and nearby farms. In situations where physical relocation was not practical or immediate evacuation, alternative mitigation measures were implemented. These included the installation of high-sensitivity fixed H_2S detectors and the structural upgrading of existing buildings for example, converting residential structures into gas-tight shelters to prevent toxic gas ingress in the event of a release. Such upgrades are critical in meeting safety thresholds and in securing approval from regulatory agencies.

It's important to note that construction works or the introduction of hydrocarbons into the system are strictly prohibited until all NOC conditions have been satisfied and verified by the relevant authorities. This includes completion of all required mitigation measures and documented compliance with environmental and safety guidelines.

Given the complexity and potential for unanticipated obstacles in this process, the timely acquisition of NOCs was placed on the project's critical path. Any delay in permitting would have had a direct impact on construction mobilization and the overall schedule.

Remarkably, through proactive planning and early engagement, the project team succeeded in securing the necessary HSEIA approvals in just four months. Furthermore, the Environment Agency granted approval to commence construction activities within one month of submission an unusually fast turnaround that reflected the completeness and quality of the pre-submittal review process. This achievement was the result of early technical due diligence, strategic stakeholder engagement, and a well-structured regulatory interface strategy. These efforts ultimately preserved the project schedule and enabled a smooth transition from planning to execution.

Strategic Procurement Acceleration and Risk Mitigation in DBO Model Execution

A major differentiator of the UC Onshore project under the DBO (Design-Build-Operate) model was the way procurement challenges were mitigated through early planning, delegation, and strategic expediting. In conventional EPC (Engineering, Procurement, and Construction) contracts, procurement of key equipment such as pressure vessels or compressors typically requires soliciting a minimum of three vendor proposals. This process entails a series of technical and commercial evaluations, vendor clarification meetings, alignment discussions, and approval loops. The technical compliance process alone involves significant coordination among vendors, engineering consultants, and the project management team. Even under best-case scenarios, the selection of a vendor and issuance of a final award recommendation could stretch to three months for medium equipment, and up to four months or more for large, long-lead items such as gas compressors.

This typical procurement timeline was significantly compressed in the UC Onshore project due to the unique structure of the DBO contract. During the tendering phase, all bidders were required to pre-nominate and submit a full list of their equipment vendors, complete with technical compliance matrices. Since the DBO contractor assumed full accountability for equipment performance and compliance, the evaluation and selection of vendors were conducted upfront, thus eliminating the need for IOC's project management team to be involved in post-award procurement technical reviews. Furthermore, the contract only mandated adherence to internationally recognized standards providing flexibility for a broader pool of vendors without compromising on quality or reliability. Figure 3 shows the difference between the DBO and EPC for different packages from date of order placement till ready at site date (ROS).

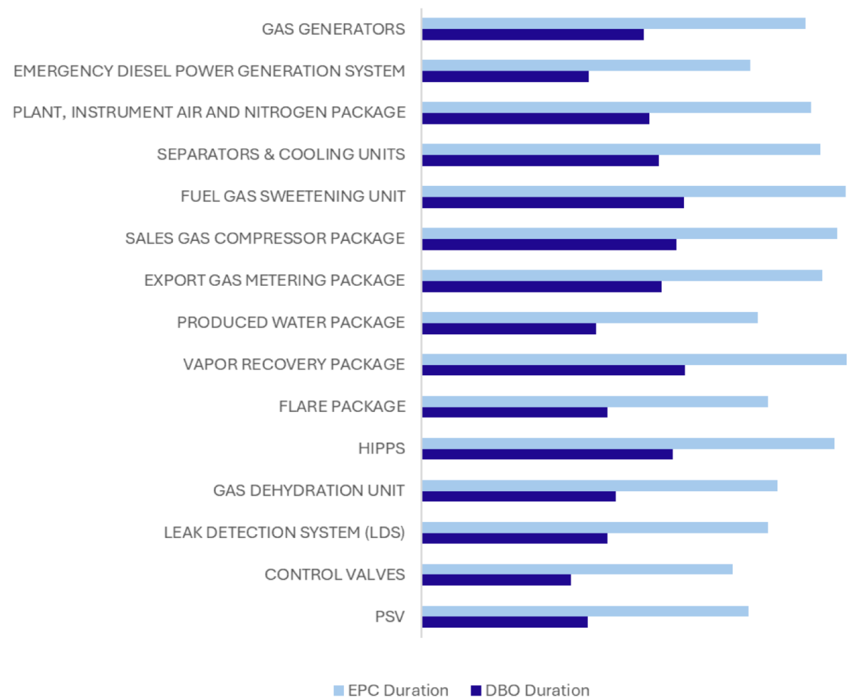

Figure 3—Comparison of Procurement, Manufacturing, and Delivery Durations in EPC and DBO

As a result, procurement activities in the UC Onshore project achieved significant acceleration. Most major purchase orders were placed within just four weeks of the project commencement, a feat rarely attainable in EPC frameworks. However, this gain necessitated a rigorous expediting regime to ensure on-time delivery, document approvals, and inspections across geographically dispersed vendor locations.

To sustain procurement velocity, a micro-level schedule was established to track and manage each step of the procurement lifecycle from vendor document submissions and review cycles to shipping, customs clearance, and on-site delivery. The UC Onshore team focused their expediting strategy on three core pillars:

1. Proactive Expediting at Vendor Premises: The team conducted repeated site visits to manufacturing yards and vendor workshops to anticipate bottlenecks, resolve fabrication issues on the ground, and ensure compliance with specifications. This proactive approach prevented delays rather than reacting to them after the fact.
2. Fast-Tracked Document Review and Approval Cycles: A strict discipline was enforced where vendor documents, inspection reports, and technical drawings were reviewed and commented on in a single round, with the second issue being treated as the final approval. This helped avoid elongated review loops and reduced turnaround time.
3. Early Risk Identification and provide mitigation plans upfront: Particularly for complex, multi-location components such as the project's main gas compressor which was manufactured in the U.S., skid-packaged in China, and finally delivered and installed in the UAE risks were pre-identified and logistics were tightly coordinated. Customs pre-clearance, multi-agency communication, and synchronized shipping schedules were essential for on-time completion.

A critical success factor was the preparation of a comprehensive critical path schedule that highlighted key dependencies and potential "blind spots" which were not captured in the initial project planning. By incorporating procurement timelines, inspection requirements, and delivery windows into the project critical path, the team was able to proactively allocate resources and avoid cascading delays.

Results and Conclusions

The UC Onshore Gas Development Project marks a milestone in IOC's journey to unlock unconventional gas, setting a new standard for fast-track delivery. Through the adoption of a Design-Build-Operate (DBO) model, the project progressed from concept to first gas in just nine months an unprecedented achievement driven by early contractor involvement, proactive planning, and flexible execution.

Key lessons emerged from this success. Early integration of the contractor allowed simultaneous engineering, procurement, and construction, minimizing interface delays. Prequalification of vendor lists during tendering enabled immediate purchase orders, drastically reducing procurement timelines. A critical path schedule tailored to long-lead items, along with proactive expediting and interface management, ensured equipment was delivered and installed without delay.

The project demonstrated that compressing all development phases into a single accountable entity can deliver exceptional results when supported by rigorous planning and clear alignment. The use of shelf equipment, flexible design scopes, and real-time coordination with stakeholders were also instrumental.

In summary, the UC Onshore project exemplifies execution excellence under a fit-for-purpose contracting strategy. It provides IOC and the industry with a scalable model for delivering complex, unconventional gas assets with speed, cost-efficiency, and operational readiness at the core.